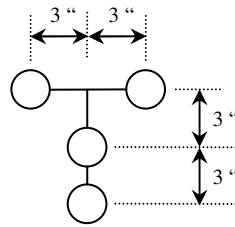


Problem #1 (15-points)

A non-standard conductor bundle for a transmission line conductor uses four ACSR *Flamingo* cables as shown below. Calculate the **geometric mean radius in feet** of this bundled conductor.



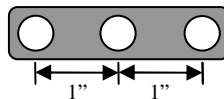
Code Name	Strands (Al/St)	Al Area (cmil)	diameter (in)	radius (ft)	GMR (ft)	R (Ω/mile) 60Hz, 20°C	ΔR/ΔT	Rated Amps
Flamingo	24/7	666600	1.000	0.0417	0.0355	0.1342	0.00054	845

Problem #2 (25-points)

A 32 mile single phase transmission line has a propagation constant of $0.0039\angle 73.26^\circ/\text{mile}$ and a characteristic impedance of $528.57\angle -9.87^\circ\Omega$. Calculate the **medium length model and long length model ABCD parameters** for this transmission line. Your final results **must** show and be accurate to **6 decimal places in magnitudes** and **3 decimal places in phase angles**. Numbers smaller than 0.001 must use scientific notation.

Problem #3 (25-points)

In the three phase power system for a small industrial facility, a generator supplies power to three distribution panels that are located in different parts of the facility and are connected to the generator through a network of power cables. The first distribution panel connects to the second distribution panel through a 450ft power cable. The first distribution panel also connects to the third distribution panel through a 750ft power cable. The second distribution panel connects to the third distribution panel through a 500ft power cable. The generator connects to first distribution panel through a 100ft power cable. The generator is Y-connected, is rated 500KVA, 480V with 21% reactance and an X/R ratio of 15, and operates with a no-load voltage of 515V. All of the facility loads are connected to the three distribution panels. The total load connected to each distribution panel is assumed to be the same at each panel and modeled by a Δ -connected load with per phase impedance of $12\angle 37^\circ\Omega$ at each panel. The power cables used in the system are all the same type, consisting of three solid #4 copper wires encased in a polyvinyl insulating material as shown below. Each #4 copper wire has a 0.127" radius and resistance of $1.518\Omega/\text{mile}$. The cable insulation has an inductance constant that is 1.03 times that of air. Using the series impedance model for all power cables, **perform PSA steps 1, 2, and 3** for this power system.



Problem #4 (35-points)

In a single phase power system, an ideal 5KV source supplies power to two parallel connected loads through a 2 mile transmission line. The transmission line uses a single ACSR *Flamingo* cable in each conductor with 3ft spacing between the two conductors. See Problem #1 for ACSR *Flamingo* cable table data. The two loads have equivalent impedances of $300\angle -35^\circ\Omega$ and a $200\angle 45^\circ\Omega$. Using **5-step PSA**, calculate the **real & reactive power supplied by the source** and the **voltage at the loads**. Use the pi model for the transmission line, and use **your choice** of equivalent circuit modeling or 2-port network modeling for the transmission line.